



# Functional Reliability of Human Digital Twins in Simulated Market Environments

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*An empirical study comparing real, synthetic, and generic digital twins across 20 market segments in B2C and B2B contexts*



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## Abstract

Human Digital Twins (HDTs) are generative agents designed to simulate human behavior in market research and innovation contexts. Despite growing adoption, empirical evidence on their reliability remains scarce, and the widely cited 85% accuracy figure from Park et al. (2024) is frequently invoked without adequate understanding of its meaning, scope, or limitations.

This study introduces the concept of functional coherence as the appropriate measure of HDT reliability for market simulation purposes. Rather than assessing whether a digital twin reproduces the same factual answers as its human counterpart, we measure whether it shares the same needs, values, behavioral patterns, and decision-making criteria — the functional space that determines how a person would react to a product, service, or value proposition.

We evaluated three categories of digital twins across 20 market segments (10 B2C, 10 B2B), each populated by 10 real individuals: Real Digital Twins built from structured interview data, Synthetic Digital Twins modeled through Aipermind’s constraint-based architecture, and Generic Digital Twins produced by instructing a general-purpose LLM to role-play a matching profile. Each individual was represented by 5 twins per category, evaluated independently by two LLMs from different families.

Real Digital Twins achieved a mean functional coherence of 86% (range: 60–92%). Aipermind Synthetic Twins achieved 78% (range: 50–97%). Generic Twins achieved 68% (range: 42–92%). The gap between synthetic and generic twins was modulated by two factors: the number of independent variables defining the segment, and the distribution of real individuals around the segment centroid. When profiles were richly specified and individuals clustered near the centroid, the gap narrowed or occasionally reversed. When profiles were sparse and individuals were distributed away from the centroid, Aipermind’s constraint-based approach showed a substantial advantage.

We argue that these findings reframe the value proposition of HDT platforms: reliability resides not primarily in the quality of individual twins, but in the architecture of the simulation process — how data is extracted, how twins are orchestrated, and how adversarial scenarios are constructed.



## 1. Introduction

### 1.1 The problem of trust

The ability to create digital representations of human individuals using large language models has expanded rapidly since 2023. Companies and researchers now routinely generate synthetic populations for market testing, product development, and policy simulation. Yet a fundamental question persists: can we trust these entities to behave as real people would?

The most commonly cited evidence is the 85% normalized accuracy reported by Park et al. (2024) in their study of 1,052 generative agents built from two-hour voice interviews. This figure has become a de facto industry benchmark, frequently invoked in marketing materials and investor presentations. However, it is rarely contextualized: 85% is a normalized score against human self-consistency (81.25% at two weeks on the General Social Survey), measured on closed-ended attitudinal questions. It does not directly address the reliability of digital twins for open-ended qualitative investigation, nor does it assess synthetic twins built from demographic profiles rather than interviews.

The gap between this single data point and the claims made by HDT vendors creates a credibility deficit that inhibits adoption. Organizations that could benefit from simulated markets hesitate because they lack evidence calibrated to their specific use case. This study aims to narrow that gap.

### 1.2 What reliability means for market simulation

In a market simulation context, digital twins serve a specific purpose: they participate in adversarial tests of value propositions, business models, and product concepts. The simulation succeeds when the twin's feedback predicts how a real person in the same market segment would react. The simulation fails when the twin provides feedback that would lead to different strategic decisions than those warranted by real-world data.

This purpose determines what we need to measure. We do not need the twin to reproduce the exact words, facts, or experiences of a real person. We need it to share the same functional space: the same needs, problems, values, behavioral patterns, and decision-making criteria. A twin that knows “cooking is a shared family activity” is functionally adequate even if it does not know the specific dishes the family prepares. Conversely, a twin that correctly names every dish but mischaracterizes the family's values around food is functionally unreliable despite its factual accuracy.

We call this measure functional coherence, and it is the central metric of this study.

### 1.3 Why questionnaires are insufficient

A straightforward approach would be to administer a questionnaire to both the real person and the twin and compare answers. Over sixty years of qualitative-quantitative research methodology demonstrate why this is inadequate. Questionnaires elicit responses shaped by social desirability bias, acquiescence bias, satisficing, and a host of other systematic distortions. The baseline they provide is not the person's true functional profile but an



artifactual representation — what we might call a counterfeit copy. Using that copy as the ground truth for twin validation would mean measuring how well a synthetic artifact matches a different synthetic artifact, with no guarantee that either reflects reality.

For this reason, our study relies on structured conversational interviews designed to minimize involuntary deception, following established principles from qualitative intelligence methods. This choice has a cost in scalability but a gain in validity: the baseline we compare against is as close to the person's authentic functional profile as current methods allow.



## 2. Theoretical framework

### 2.1 Functional coherence

We define functional coherence as the degree to which a digital twin shares the real person's functional space along five dimensions:

- **Needs:** the practical, emotional, and relational requirements the person experiences in the domain under investigation.
- **Problems:** the obstacles, frictions, and pain points the person encounters in addressing those needs.
- **Values:** the criteria the person uses to judge what matters and what does not.
- **Behavioral patterns:** the recurring habits, routines, and ways of acting that characterize the person's approach.
- **Decision-making criteria:** the way the person weighs alternatives when facing a choice.

Functional coherence is deliberately agnostic to factual accuracy. A twin may invent biographical details, name different products, or describe different episodes. As long as the invented content expresses the same functional profile, it is coherent. This distinction is not merely methodological: it reflects the actual use case. When a product manager tests a concept against a digital twin, the question is not “would this specific person buy it?” but “would someone with these needs, values, and decision patterns buy it?”

### 2.2 The role of constraints in identity formation

A central theoretical observation motivates our approach to synthetic twin construction. Human identity is not a list of attributes but the specific shape that emerges from the intersection of mutually constraining events, experiences, and circumstances. Being a mother constrains your schedule; your schedule constrains your cooking habits; your cooking habits constrain your relationship with food; your relationship with food constrains your reactions to food products. No single constraint defines you, but their particular configuration does.

A general-purpose LLM instructed to “behave as if” it were a person of a given profile lacks this internal constraint network. Each response is generated independently from the same probability distribution over the profile's attribute space. The result is a statistically plausible but structurally flat persona — a centroid of all possible people matching the profile, rather than a specific individual within that space.

Aipermind's synthetic twin architecture addresses this by constructing synthetic memories: a sequence of mutually coherent biographical events, experiences, and circumstances that create internal constraints. Each memory narrows the space of plausible subsequent responses. The resulting twin is not a real person, but it has the structural properties of a real person: internal coherence, specificity, and resistance to collapse toward the statistical mean.



This mechanism predicts that the advantage of constraint-based twins over generic twins should be inversely proportional to the richness of the external profile. When the profile provides many specific attributes, external constraints substitute for internal ones, and both approaches converge. When the profile is sparse, only internal constraints prevent the twin from defaulting to the segment stereotype. Our data confirm this prediction.

### 2.3 Market segments and the centroid problem

A market segment is a group of individuals equally sensitive to a value proposition and therefore equally exposed to the problem that the proposition addresses. It is defined by independent variables (demographic, behavioral, psychographic attributes that can be observed and measured) and dependent variables (needs, problems, decision criteria that causally follow from the independent variables). The independent variables constitute a profile; the dependent variables constitute the functional space we aim to reproduce.

The centroid of a segment is a geometric concept: the average position across all independent variables. Its relationship to actual individuals in the segment varies. In some segments, many individuals cluster near the centroid: the segment is homogeneous, and the centroid is a reasonable proxy for a typical member. In others, individuals are distributed away from the centroid, which may correspond to no one in particular: the segment is heterogeneous, and the centroid is a statistical fiction.

This distinction is critical for twin validation. A generic LLM, lacking internal constraints, tends to produce responses corresponding to the segment centroid. When real individuals cluster near the centroid, this tendency is an advantage: the generic twin matches well. When real individuals are distributed away from the centroid, the generic twin matches poorly because it produces the behavior of a person who does not exist. Aipermind's constraint-based twins, by generating specific individuals rather than centroid approximations, maintain reliability across both configurations.



## 3. Experimental method

### 3.1 Design overview

The experiment compared four entities for each of 200 real individuals (100 B2C, 100 B2B) distributed across 20 market segments:

| Entity                 | Source                       | Abbreviation | N per individual |
|------------------------|------------------------------|--------------|------------------|
| Real person            | Structured interview         | A            | 1                |
| Real Digital Twin      | Interview minus 20% holdout  | ARDT         | 5                |
| Synthetic Digital Twin | Aipermind constraint-based   | ADT          | 5                |
| Generic Digital Twin   | General-purpose LLM prompted | AGT          | 5 per model (×2) |

### 3.2 Population and segments

We selected 10 B2C and 10 B2B segments across diverse industries. The distinction is substantive: B2C segments are defined primarily by personal demographic and behavioral variables, while B2B segments incorporate professional role, organizational context, and industry-specific variables. Each segment was populated by 10 real individuals who had been previously validated as segment members through independent variable verification.

Segments were deliberately selected to vary on two dimensions: the number of independent variables defining the profile (from minimal profiles with 3–4 variables to rich profiles with 8+ variables), and the distribution of real individuals around the segment centroid (from concentrated distributions where most individuals cluster near the center to dispersed distributions where individuals occupy the periphery).

### 3.3 Baseline collection

Each real individual participated in a structured conversational interview designed to elicit authentic responses about their needs, problems, behaviors, and decision-making processes in the domain relevant to their segment. The interview protocol followed established qualitative intelligence principles to minimize involuntary deception: open-ended questions, progressive depth, absence of leading prompts, and validation through behavioral anchoring (asking about specific recent experiences rather than abstract preferences).

For the holdout test of Real Digital Twins, 20% of the question-answer pairs were removed from the transcript before twin construction. The removed pairs were selected to cover diverse topics within the interview.

### 3.4 Twin construction

#### Real Digital Twins (ARDT)



Built from the interview transcript minus the holdout set, using the same modeling stack employed in Aipermind's production pipeline. The twin receives the reduced transcript as its biographical grounding. Five instances were generated per individual to measure intra-twin variance.

### **Synthetic Digital Twins (ADT)**

Built through Aipermind's constraint-based architecture, which constructs synthetic memories from the independent variables of the individual's segment profile. The process generates a sequence of mutually coherent biographical events that create internal constraints, producing a specific individual rather than a centroid approximation. Five instances were generated per individual.

### **Generic Digital Twins (AGT)**

Built by instructing a general-purpose LLM to respond as if it were a person matching the same independent variable profile. Two different LLM families were used, neither of which corresponds to the LLM used for Aipermind's twin construction. Five instances were generated per individual per model.

## **3.5 Interview protocol for twins**

All twins were subjected to the same conversational interview as the real person, with one critical modification: follow-up questions from the original interview were removed to ensure reproducibility and prevent implicit nudging. The synthetic interviewer was designed to avoid biased or leading questions. The same questions, in the same order, were administered to all twin types.

## **3.6 Evaluation**

Functional coherence was assessed by two LLMs from different families, both distinct from the LLM used for twin generation. Each evaluator independently analyzed every question-answer pair by extracting functional propositions from both the baseline and twin responses, classifying each proposition as functional (expressing a need, value, pattern, criterion, or preference) or accessory (expressing a specific factual detail), matching functional propositions between baseline and twin, and assigning a coherence degree (identical, coherent, contrary, or absent).

Functional coherence was calculated as the proportion of baseline functional propositions that found an identical or coherent match in the twin's response. The final score for each pair was the mean of the two evaluators' assessments. Inter-evaluator variance was measured and found to be low across all conditions, confirming that the two evaluations can be considered equivalent.

## **3.7 What we did not measure**

To maintain focus on functional coherence, we deliberately excluded from measurement: literal correspondence between responses, verbal density and response length, tone and emotional register, specific expressive forms, idiosyncratic detail correspondence, and any





other dimension not included in the functional coherence framework. These exclusions are not oversights but design choices: they reflect the principle that market simulation reliability depends on functional alignment, not surface similarity.



## 4. Results

### 4.1 Overall functional coherence

| Twin type       | Mean | Range  | Interpretation                                   |
|-----------------|------|--------|--------------------------------------------------|
| Real (ARDT)     | 86%  | 60–92% | Baseline: maximum achievable with interview data |
| Synthetic (ADT) | 78%  | 50–97% | Constraint-based approach from profile           |
| Generic (AGT)   | 68%  | 42–92% | Prompt-based approach from same profile          |

The 86% functional coherence of Real Digital Twins represents the empirical ceiling for this experimental design. It indicates that even with full access to the interview data (minus the holdout), the twin’s responses do not perfectly replicate the real person’s functional profile. The residual 14% gap reflects the irreducible distance between a generative model and a real individual, including aspects of functional identity that emerge only in specific conversational contexts and cannot be inferred from prior data.

### 4.2 B2B vs B2C

Functional coherence was consistently slightly higher in B2B segments than in B2C segments, regardless of the twin construction method. This finding likely reflects the more constrained nature of professional roles: a B2B individual’s needs and decision criteria are substantially shaped by organizational context, industry norms, and role-specific responsibilities, all of which are more predictable from independent variables than the personal lifestyle factors that drive B2C behavior.

### 4.3 Effect of profile richness

In segments defined by few independent variables (3–4 attributes), all twin types showed lower functional coherence than in richly defined segments (8+ attributes). However, the effect was not uniform across twin types. Generic twins showed the steepest decline with sparse profiles, while Aipermind’s synthetic twins degraded more gradually, and Real Digital Twins were least affected. This pattern is consistent with the theoretical prediction that internal constraints (generated by the memory construction process) compensate for the absence of external constraints (provided by the profile).

### 4.4 Effect of individual distribution around the centroid

This is the most consequential finding. In segments where real individuals clustered near the centroid, the gap between Aipermind’s synthetic twins and generic twins was small, and in some cases reversed (generic twins outperformed synthetic twins). In segments where real individuals were distributed away from the centroid, the gap widened substantially in favor of Aipermind’s approach.



The interpretation is straightforward. A generic LLM, lacking internal constraints, generates responses that correspond to the segment's statistical center. When real individuals happen to be near that center, this is an effective strategy. When they are not, the generic twin produces the functional profile of a person who does not exist in the segment.

Aipermind's constraint-based twins, by generating specific individuals with internal coherence, produce functional profiles that are dispersed across the segment space rather than concentrated at the centroid. This dispersion better matches the actual distribution of real individuals in heterogeneous segments.

#### 4.5 Interaction effects

The two moderating variables (profile richness and centroid distribution) interact. The maximum advantage of constraint-based twins over generic twins occurs in segments that are both sparsely defined and heterogeneously distributed. The minimum advantage (or reversal) occurs in segments that are richly defined and homogeneously distributed. These are the two extremes of a continuum along which the relative value of internal constraints varies.

The difference between the extremes of the overall distribution (42–97% for synthetic, 42–92% for generic) is driven not by industry sector but by these two structural properties of the segment: how many variables define it, and how its members are distributed around the centroid.



## 5. Interpretation

### 5.1 Why internal constraints matter

Human beings are the particular result of a series of constraints determined by a series of events that constrain each other. The specific shape assumed by this constraint network constitutes individual identity. A person is not defined by any single attribute but by the mutual configuration of all their attributes, experiences, and circumstances.

When we construct a digital twin by instructing an LLM to “behave as if” it were a person matching a profile, we provide external constraints only. The LLM fills in the gaps by sampling from the most probable distribution given those constraints. The result is not a specific individual but the centroid of the space of all individuals compatible with the profile — a statistical fiction.

Aipermind’s constraint-based architecture replicates the mechanism through which real identity forms: it generates sequences of mutually coherent biographical events that progressively narrow the space of plausible responses. The first synthetic memory constrains the second; the second constrains the third; each layer reduces the space of subsequent possibilities. The resulting twin is not a real person, but it has the structural properties of one: internal coherence, specificity, and resistance to collapse toward the statistical mean.

This mechanism explains why constraint-based twins show their greatest advantage precisely when external constraints are insufficient: sparse profiles and dispersed segments. In these conditions, the LLM’s natural tendency to generalize is unchecked by external data, and only internal constraints prevent the twin from defaulting to the stereotype. Conversely, when external constraints are abundant (rich profiles) or when the centroid happens to match reality (homogeneous segments), the advantage diminishes because external constraints alone are sufficient to prevent excessive generalization.

### 5.2 The centroid is not a person

An important objection is that the segment centroid, being the most statistically representative point, should be the optimal target for product design. If designing for the average person in the segment, a generic twin that naturally produces centroid-like responses would be preferable.

This objection holds when the distribution of real individuals is symmetric and unimodal around the centroid, the product is an incremental improvement to an existing category, and the relevant metric is average acceptance rather than distribution of reactions. It fails in at least three common scenarios: when the segment contains distinct sub-clusters (the centroid falls between them, corresponding to no one), when business value resides in the distribution tails (churn, advocacy, early adoption), and when the product is genuinely new and the relevant question is whether a viable segment exists at all, not how the known average reacts.

In market simulation for innovation testing — the primary use case for Aipermind — these scenarios are the norm rather than the exception. Testing a new business model requires



exposure to the variability of human reactions, not convergence on the average reaction. A simulated market composed of centroid-approximate individuals will consistently overestimate product acceptance within the target segment and fail to surface the friction points, edge cases, and unexpected objections that determine real-world outcomes.

### 5.3 The analogy with traditional market research

The relationship between twin quality and process quality mirrors the history of traditional market research. No research institute ever sold “better respondents.” The value was always in the architecture of the research process: how the sample was constructed, how questions were formulated, how biases were controlled, and how results were interpreted. An Ipsos respondent was not “better” than a Nielsen respondent; the difference lay in the system.

Similarly, our data suggest that individual twin quality is approaching commodity status. Any competent LLM with a reasonable profile can produce twins with 60–80% functional coherence. The differentiating value lies not in the twin but in the simulation environment: how data is extracted, how twins are orchestrated within a market architecture, how adversarial scenarios are constructed, and how feedback is interpreted and translated into actionable insights.

This reframing has practical implications. The question for organizations evaluating HDT platforms should not be “how good are your twins?” but “how reliable is your simulation process?” A good twin subjected to a poor investigation methodology produces worse results than a mediocre twin subjected to a rigorous one. The twin is a component; the process is the product.

### 5.4 On the fragility of synthetic twins

An important caveat emerged from our testing: all synthetic twins, whether constraint-based or generic, lose reliability dramatically when independent variables are confused with dependent variables in the profile definition, or when leading questions or other forms of bias are introduced in the conversation protocol. This finding has implications beyond our study: the quality of commercially available digital twins is likely to vary enormously depending on construction methodology, and organizations using external twin providers without understanding their construction process face substantial risk of unreliable results.



## 6. Limitations

- **Scope.** We tested 20 segments across multiple industries but did not cover all possible sector-segment-profile combinations. The results should be interpreted as indicative of patterns rather than definitive benchmarks for specific domains.
- **Evaluation dependency.** Functional coherence was assessed by LLMs, not humans. While inter-evaluator agreement was high and conceptually equivalent, the evaluation remains within the LLM ecosystem. A future study should include human evaluator calibration on a representative subsample.
- **Baseline validity.** Our baseline relies on structured interviews, which are superior to questionnaires but still subject to residual biases. The “true” functional profile of a person is ultimately unobservable; our baseline is the best available approximation, not ground truth.
- **Centroid measurement.** The distribution of individuals around the segment centroid was characterized qualitatively (concentrated vs. dispersed) rather than through precise geometric metrics. Future work should formalize this dimension.
- **Temporal stability.** We measured functional coherence at a single point in time. How twin reliability evolves as real individuals’ circumstances change remains untested.



## 7. Conclusions

This study offers what we believe to be the first empirical measurement of functional coherence for digital twins evaluated against open-ended qualitative interview data across multiple market segments and twin construction methods.

We invested in constructing an experiment whose results we would trust ourselves — even at the cost of confronting uncomfortable findings. The effort was substantial, and we did not cover all possible cases in all possible sectors with all possible nuances. What we found can be summarized in five points.

1. **Functional coherence is the right metric.** For market simulation purposes, what matters is not whether a twin says the same things as a real person, but whether it shares the same functional space of needs, values, and decision criteria. Our framework provides an operational method for measuring this.
2. **Real Digital Twins achieve 86% functional coherence,** establishing an empirical ceiling for interview-based twin construction. This is broadly consistent with Park et al.'s normalized accuracy but measured on qualitative data using functional rather than factual criteria.
3. **Constraint-based synthetic twins (78%) outperform generic twins (68%) on average,** but the gap is modulated by profile richness and segment heterogeneity. The advantage is maximal in sparse, dispersed segments and minimal in rich, concentrated ones.
4. **A 68% functional coherence is already high.** We would not recommend paying a substantial premium in cost, time, or latency to move from 68% to 78%. The marginal value of twin quality improvement diminishes rapidly when compared to the value of process quality improvement. Most traditional market research operates at effective reliability levels well below 68% due to methodological shortcomings that amplify biases and involuntary deception.
5. **Reliability resides in the process, not the twin.** Even a prompt-based twin, when coupled with a sound investigation methodology and a rigorous extraction of meaning, produces higher quality insights than the best twin subjected to a poor investigation strategy. What Aipermind provides is not better twins per se, but a simulation environment in which good twins are engaged in reliable investigative processes that produce insights useful for evolving projects without creating problems in the reality of atoms.

The path forward is not to pursue marginal improvements in individual twin fidelity but to invest in the architecture of simulation: better segment definition, better scenario construction, better interpretation frameworks. The twins are good enough. The question is whether the world we build around them is.



## References

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+ The complete list of references is available on request [hello@aipermind.com](mailto:hello@aipermind.com)

